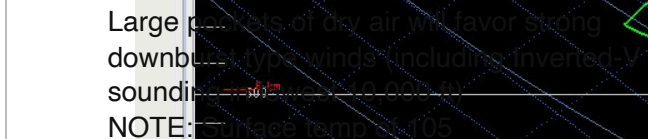


Review of August 11, 2015 Severe Weather Event

Eric Zappe
National Weather Service
Houston Center Weather Service Unit



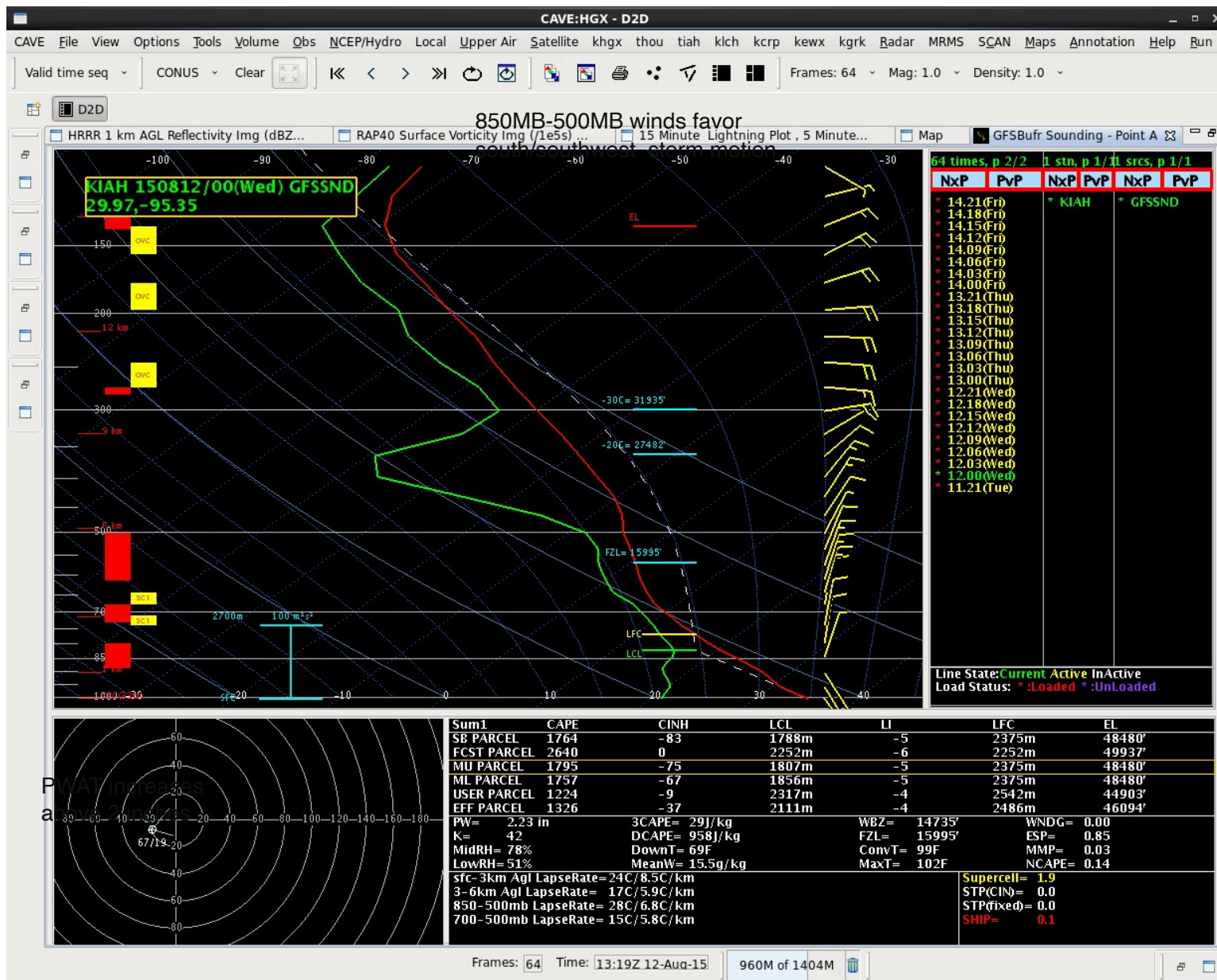
GFS BUFR Sounding – KIAH (Aug 11 21Z)



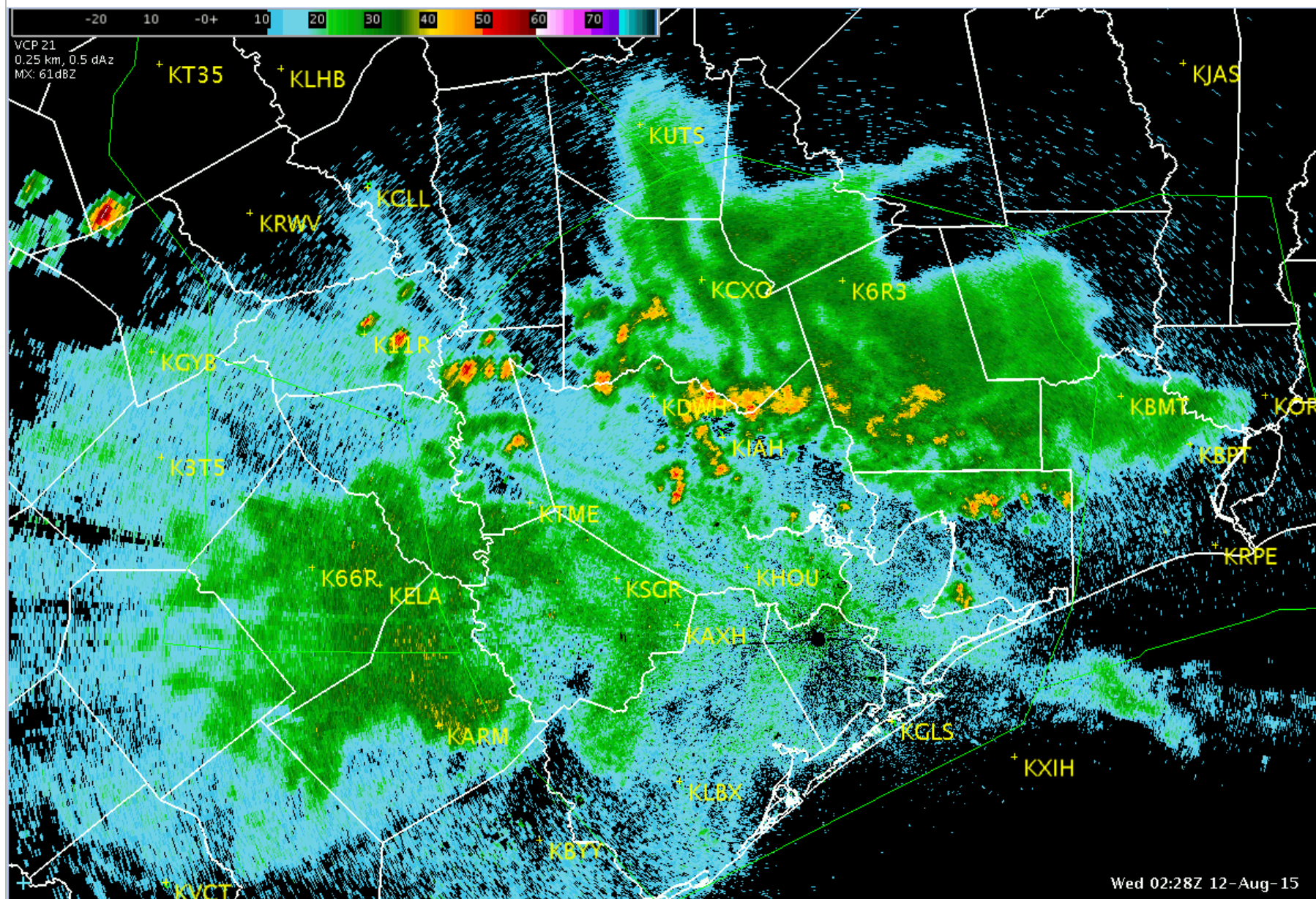
Elevated DCAF

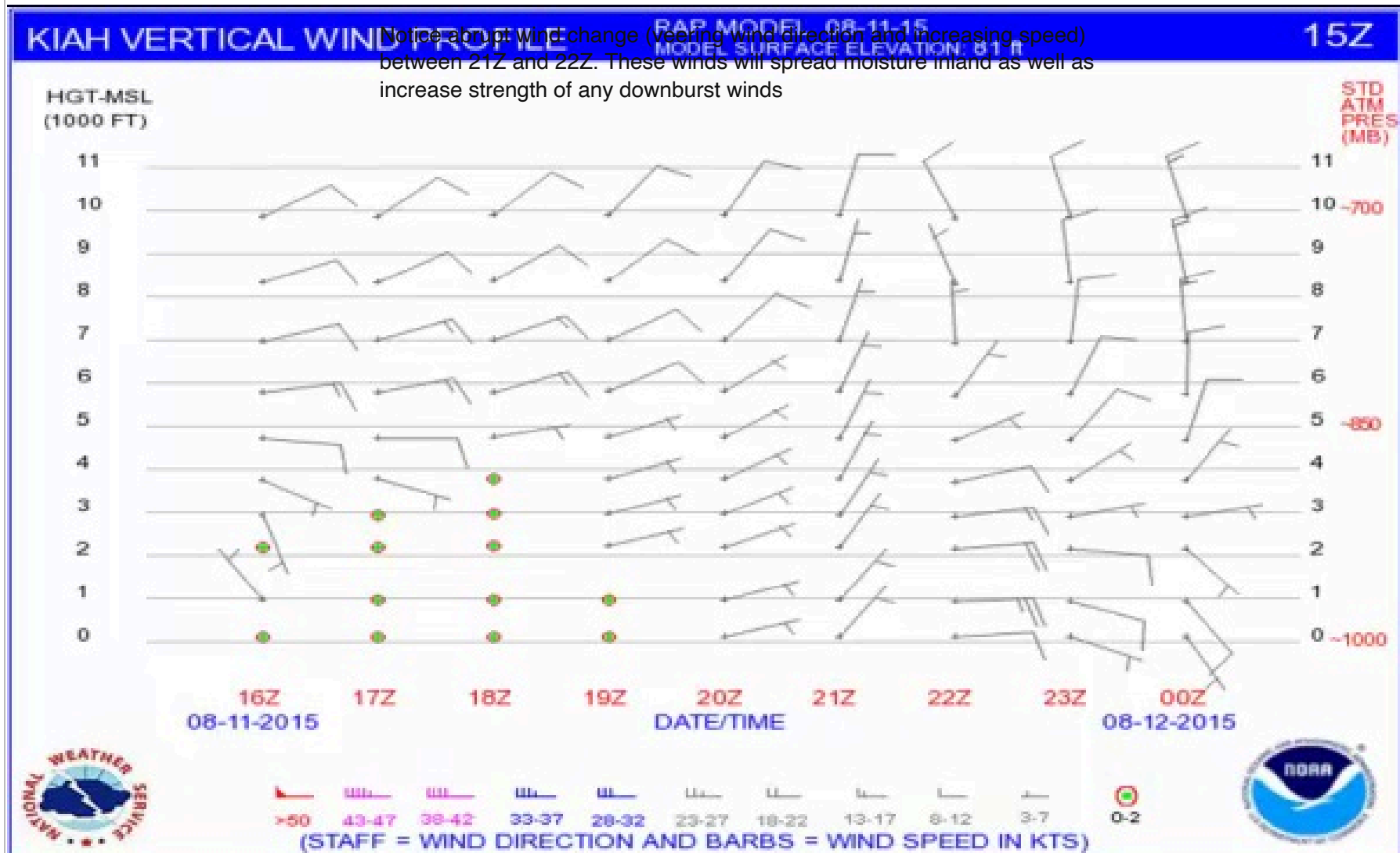
Click Here for
DCAPE Study

GFS BUFR Sounding – KIAH (Aug 12 00Z)

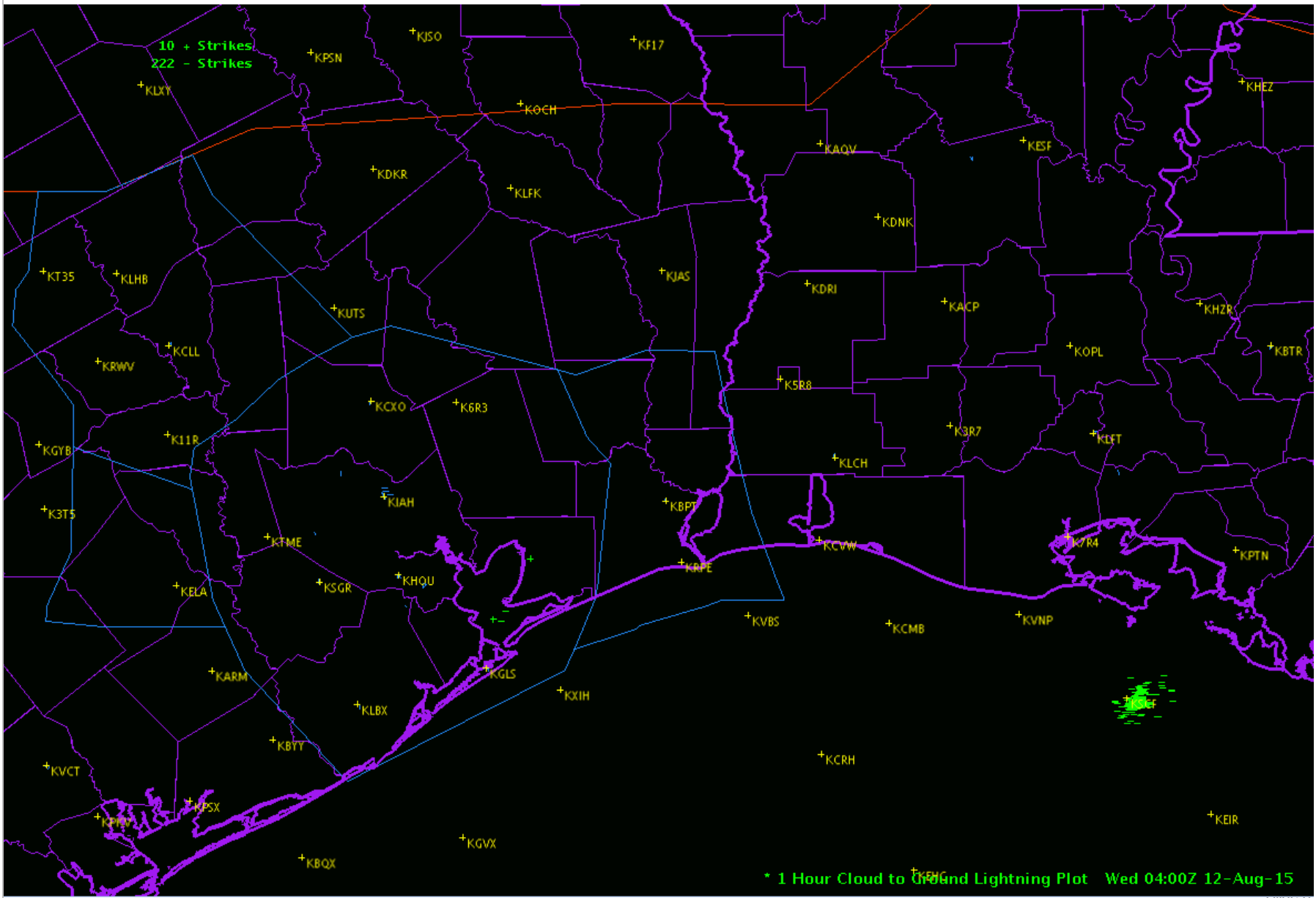


kiah Reflectivity Loop (Aug 11 20:27Z – Aug 12 02:28Z)

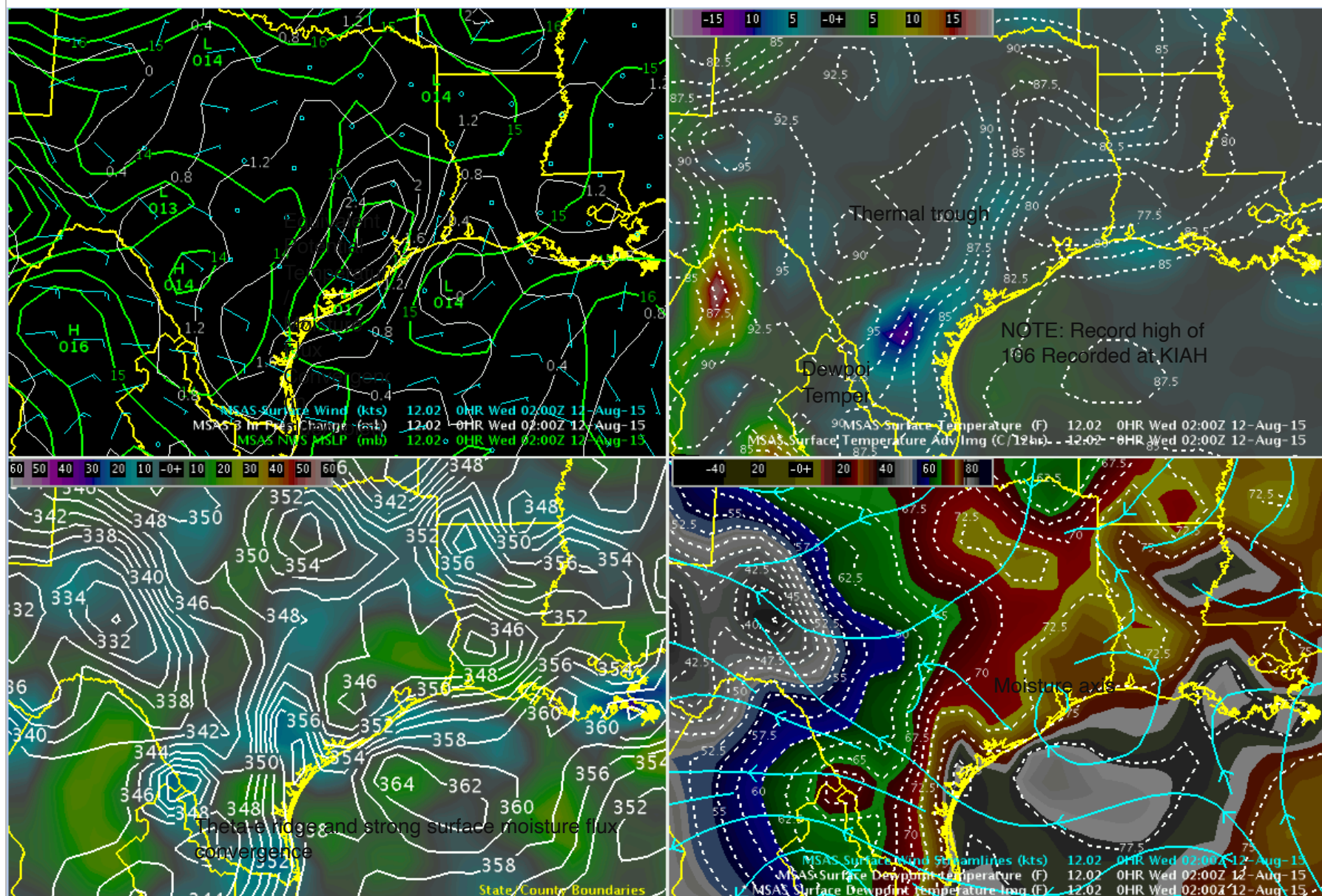


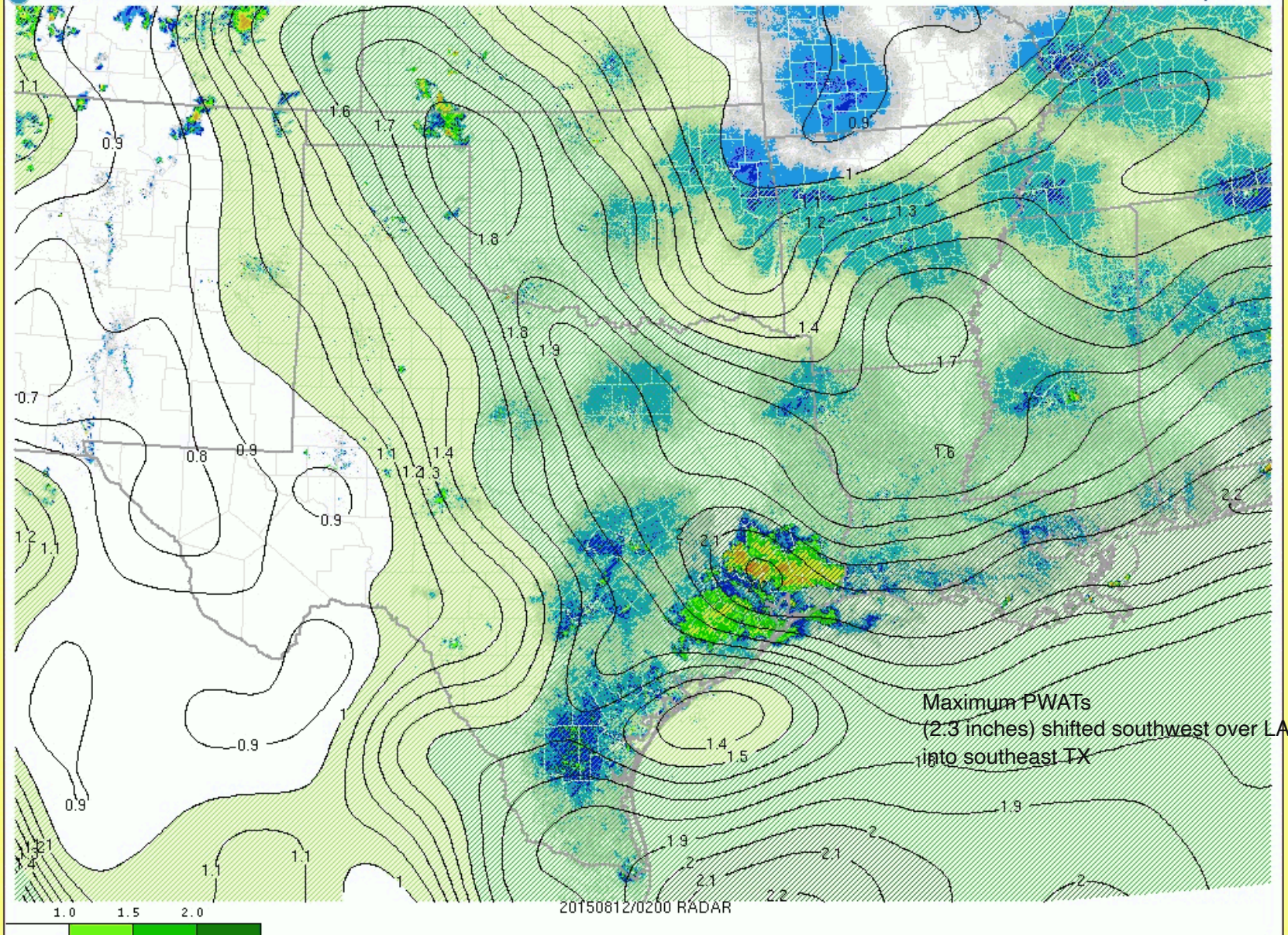


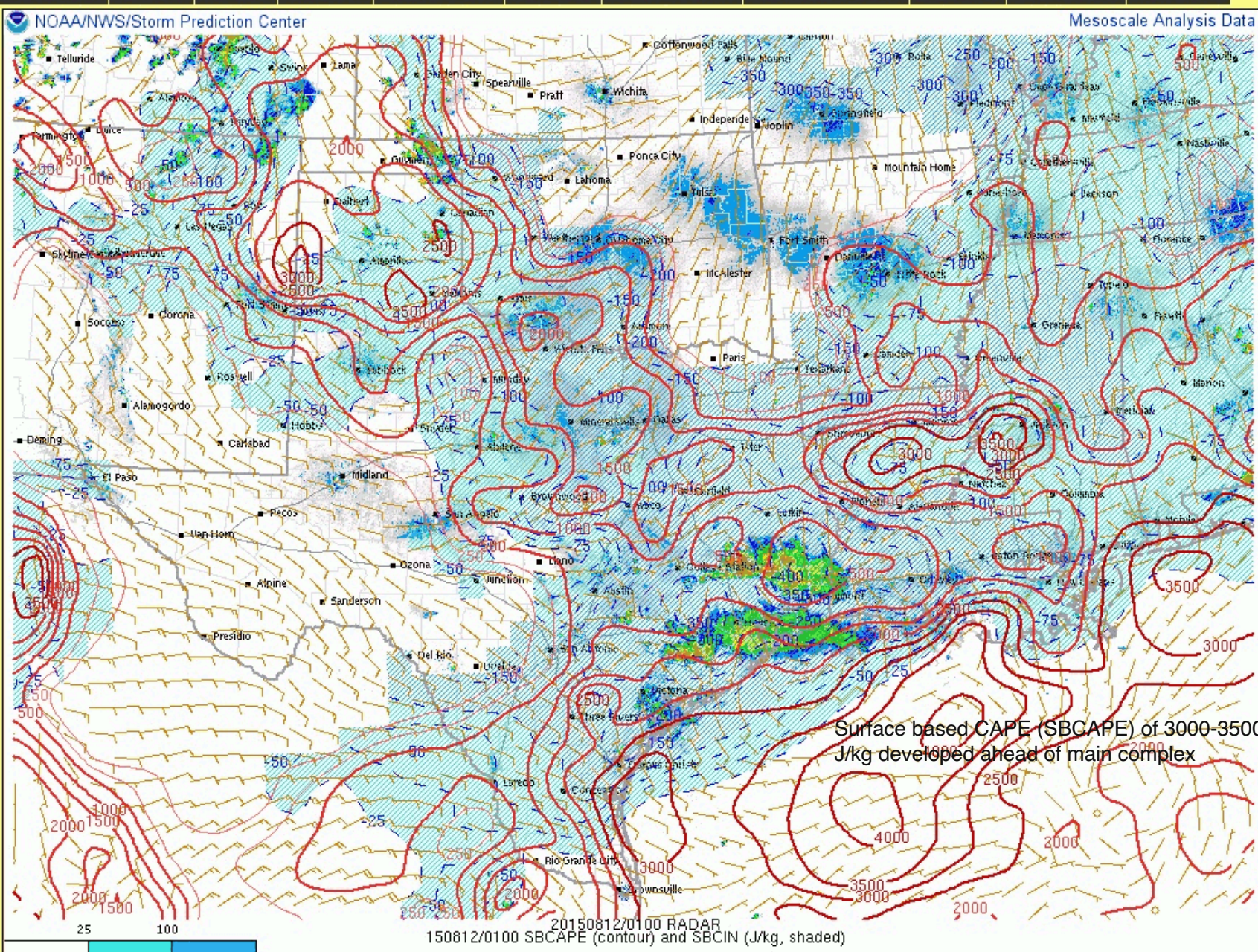
One Hour Lightning Plot Loop (Aug 11 19Z - Aug 12 02Z)



Temper





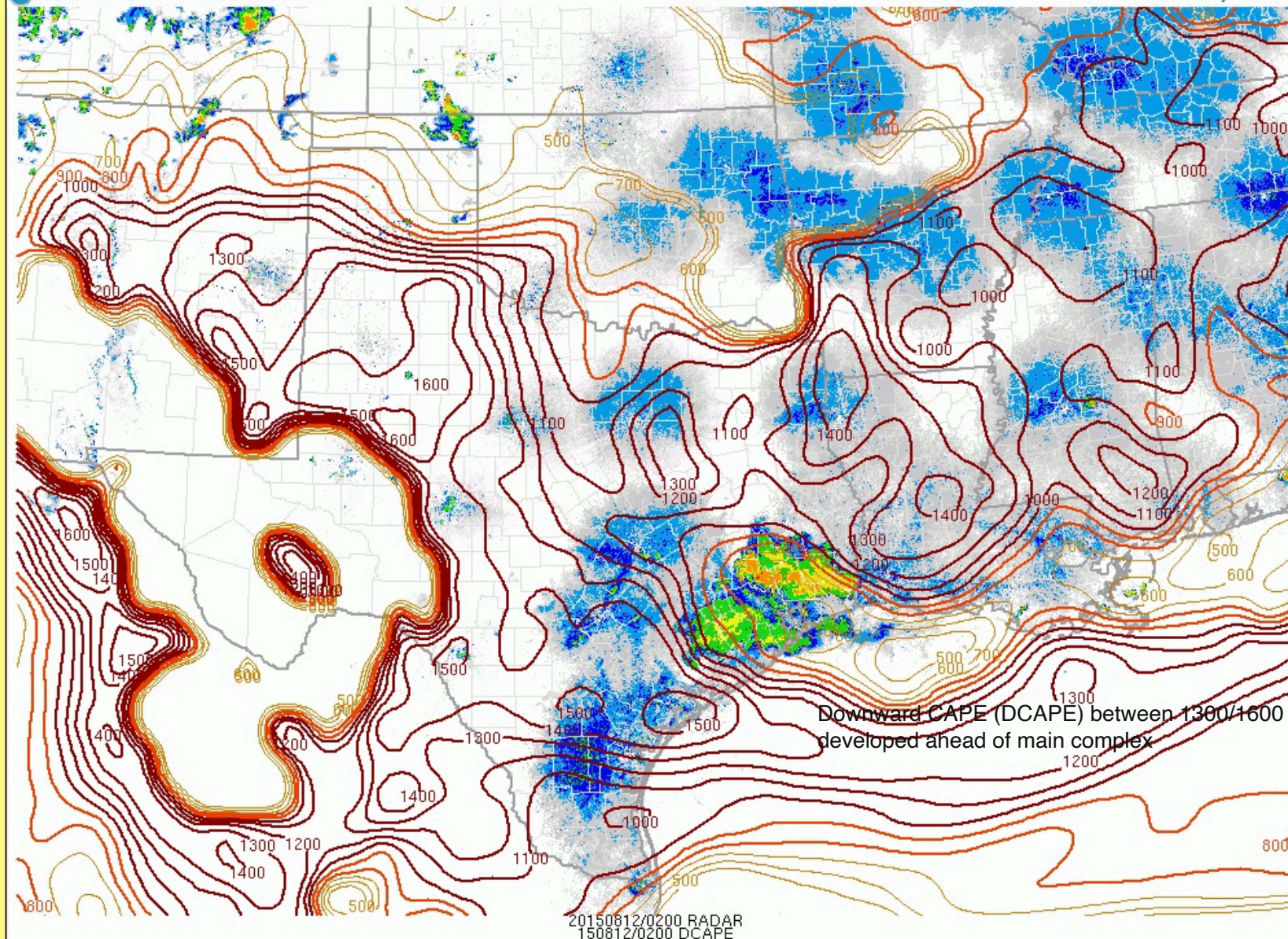


SPC Mesoscale Analysis - Archive

[Change Sector](#) [Loop Options](#) [SPC Homepage](#)[Return to Real-Time Viewer](#)[Observations](#) [Basic Sfc](#) [Basic UA](#) [Kinematics](#) [Thermodynamics](#) [Wind Shear](#) [Composite Indices](#) [Multi-Parameter Fields](#) [Heavy Rain](#) [Winter Weather](#) [Fire Weather](#)

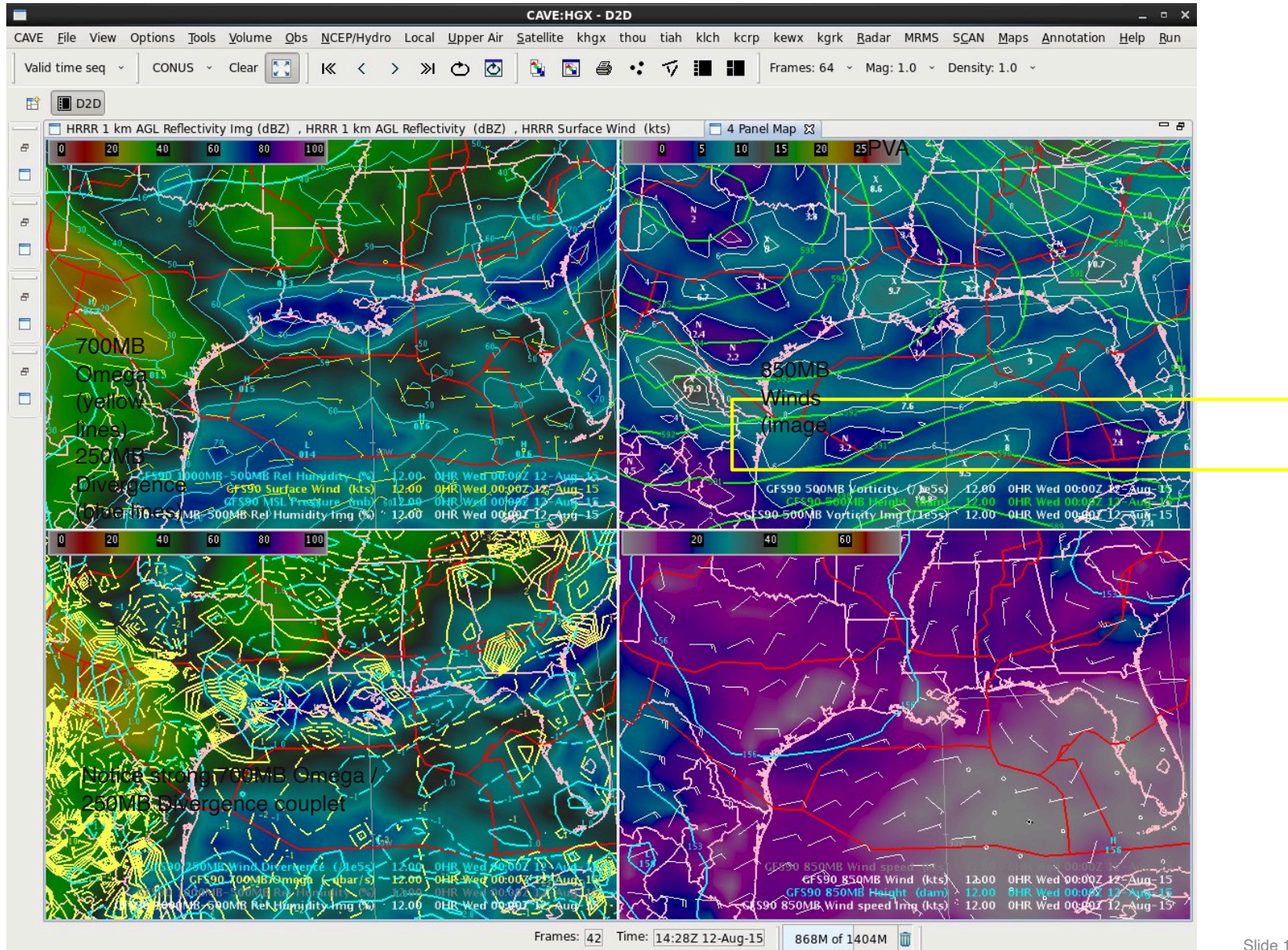
NOAA/NWS/Storm Prediction Center

Mesoscale Analysis Data



GFS90 4-Panel (Aug-12 00Z)
(image)

Vorticity
(image)



Record High Temperatures Set on Aug 11

RECORD EVENT REPORT
NATIONAL WEATHER SERVICE HOUSTON/GALVESTON TX
0234 AM CDT WED AUG 12 2015

...RECORD HIGH TEMPERATURE WAS SET AT HOUSTON INTERCONTINENTAL
AIRPORT YESTERDAY...

A RECORD HIGH TEMPERATURE OF 106 DEGREES WAS SET AT HOUSTON
INTERCONTINENTAL AIRPORT YESTERDAY. THIS BREAKS THE OLD RECORD OF
102 SET IN 2011.

...RECORD HIGH TEMPERATURE WAS SET AT HOUSTON/HOBBY AIRPORT...

A RECORD HIGH TEMPERATURE OF 104 DEGREES WAS SET AT HOUSTON/HOBBY
AIRPORT YESTERDAY. THIS BREAKS THE OLD RECORD OF 101 SET IN 2011.

...RECORD DAILY MAXIMUM RAINFALL WAS SET AT HOUSTON/HOBBY AIRPORT...

A RECORD RAINFALL OF 2.61 INCH(ES) WAS SET AT HOUSTON/HOBBY AIRPORT
YESTERDAY. THIS BREAKS THE OLD RECORD OF 1.97 SET IN 1983.

PRELIMINARY LOCAL STORM REPORT...SUMMARY
 NATIONAL WEATHER SERVICE HOUSTON/GALVESTON TX
 814 AM CDT WED AUG 12 2015

..TIME...	...EVENT...	...CITY LOCATION...	...LAT.LON...
..DATE...	MAG....	..COUNTY LOCATION..ST..	...SOURCE....
	..REMARKS..		
0424 PM 08/11/2015	TSTM WND DMG	5 WSW MONTGOMERY MONTGOMERY TX	30.36N 95.77W BROADCAST MEDIA
	TREES BLOWN DOWN... ONE ON A TRUCK		
0454 PM 08/11/2015	TSTM WND GST M60 MPH	7 S TOMBALL HARRIS TX	30.00N 95.62W ASOS
	THUNDERSTORM WIND GUST WAS MEASURED AT DWH ASOS SITE.		
0554 PM 08/11/2015	MARINE TSTM WIND M43 MPH	2 E MORGAN'S POINT HARRIS TX	29.68N 94.98W PORTS
	WIND GUST WAS MEASURED AT MORGANS POINT PORTS SITE.		
0600 PM 08/11/2015	TSTM WND DMG	HOUSTON HARRIS TX	29.77N 95.39W BROADCAST MEDIA
	FREEWAY SIGN WIND DAMAGE AT SCOTT STREET AND LOOP 610		
0600 PM 08/11/2015	TSTM WND DMG	11 ENE CALDWELL BURLESON TX	30.59N 96.53W BROADCAST MEDIA
	3 TREES DOWN NEAR HIGHWAY 21 AND BRAZOS RIVER. KBTX MEDIA RETWEET.		
0615 PM 08/11/2015	MARINE TSTM WIND M43 MPH	4 NNE SAN LEON GMZ335 TX	29.54N 94.91W MESONET
	WIND GUST WAS MEASURED AT A WEATHERFLOW SITE IN GALVESTON BAY.		
0618 PM 08/11/2015	MARINE TSTM WIND M39 MPH	13 E BAYOU VISTA GMZ355 TX	29.36N 94.73W PORTS
	WIND GUST WAS MEASURED AT GALVESTON NORTH JETTY PORTS SITE.		
0619 PM 08/11/2015	TSTM WND GST M69 MPH	3 W SOUTH HOUSTON HARRIS TX	29.65N 95.28W ASOS
	ASOS REPORT FROM HOUSTON HOBBY AIRPORT.		
0624 PM 08/11/2015	HAIL E1.00 INCH	2 NW SUGAR LAND FORT BEND TX	29.64N 95.64W PUBLIC

0631 PM	TSTM WND GST	SUGAR LAND	29.62N	95.62W
08/11/2015	M61 MPH	FORT BEND	TX	ASOS

WIND GUST WAS MEASURED AT SUGAR LAND ASOS SITE KSGR.

0642 PM	TSTM WND DMG	2 WNW SOUTH HOUSTON	29.67N	95.26W
08/11/2015		HARRIS	TX	PUBLIC

LARGE TREE WAS DOWNED IN A BACKYARD NEAR THE
INTERSECTION OF MONROE ROAD AND INTERSTATE 45.

0642 PM	HAIL	RICHMOND	29.58N	95.77W
08/11/2015	E0.88 INCH	FORT BEND	TX	PUBLIC

NICKEL SIZE HAIL

0644 PM	MARINE TSTM WIND	13 E BAYOU VISTA	29.36N	94.73W
08/11/2015	M39 MPH	GMZ355	TX	PORTS

WIND GUST WAS MEASURED AT NORTH JETTY PORTS SITE.

0645 PM	HAIL	RICHMOND	29.58N	95.77W
08/11/2015	M1.00 INCH	FORT BEND	TX	PUBLIC

QUARTER SIZE HAIL IN RICHMOND.

0647 PM	MARINE TSTM WIND	3 SE CHANNELVIEW	29.77N	95.08W
08/11/2015	M46 MPH	HARRIS	TX	MESONET

WIND GUST WAS MEASURED AT A WEATHERFLOW SITE IN
GALVESTON BAY.

0721 PM	LIGHTNING	KENDLETON	29.45N	96.00W
08/11/2015		FORT BEND	TX	LAW ENFORCEMENT

CAR SEVERELY DAMAGED BY LIGHTNING. TIME ESTIMATED.

KIAH Observations on Aug 11

Weather Conditions for:

Houston Intercontinental Airport, TX. KIAH (NWS/FAA)

Elev: 89 ft; Latitude: 29.98440; Longitude: -95.36074

NOTE: Record high of 106

Time	Temp.	Dew	Relative	Wind	Wind	Visibility	WX	Clouds	Sea Level	Altimeter	Station	Precip	Precip	Precip	6 Hr
(CDT)	(f)	(f)	(%)	Direction	Speed	(miles)			(mb)	Setting	Pressure	1 hour	3 hour	6 hour	Max
11 Aug 10:27 pm CDT	77	73	89	SE	9	8.00	-RA	FEW035 SCT075 OVC110		30.04	29.952	0.03			
11 Aug 10:13 pm CDT	77	73	89	SE	8	9.00	TSRA	SCT035CB SCT075 OVC100		30.04	29.952	0.02			
11 Aug 9:53 pm CDT	77	73	89	ESE	18	4.00	TSRA BR	SCT033CB SCT060 OVC095	1017.4	30.05	29.962	0.09	0.09		
11 Aug 9:28 pm CDT	81	73	79	NE	18	10.00	TSRA	SCT030CB SCT070 BKN100 OVC180		30.09	30.002	T			
11 Aug 9:15 pm CDT	82	72	70	NE	21	10.00	TSRA	SCT030CB SCT070 BKN110 OVC180		30.09	30.002	T			
11 Aug 9:09 pm CDT	82	72	70	NE	20	10.00		FEW030 SCT070 BKN085 BKN180 OVC250		30.08	29.992				
11 Aug 8:53 pm CDT	83	72	70	NNE	18	10.00		FEW075 SCT110 BKN180 OVC250	1017.6	30.05	29.962				
11 Aug 7:53 pm CDT	81	70	69	S	9	10.00		FEW060 SCT150 BKN180 BKN250	1015.2	29.98	29.893	T			
11 Aug 6:53 pm CDT	84	71	65	S	16	10.00	-RA	FEW035 SCT065 BKN140 BKN180 BKN250	1014.8	29.97	29.883	T		T	106
11 Aug 6:27 pm CDT	86	70	59	ESE	22	10.00		FEW030 SCT065 BKN150 BKN170 BKN250		29.96	29.873				
11 Aug 5:53 pm CDT	91	70	50	NW	3	10.00		FEW035 SCT070 BKN100 BKN150 BKN250	1013.5	29.93	29.843				
11 Aug 5:09 pm CDT	93	66	41	NNE	14	10.00		FEW040 SCT075 BKN130 BKN250		29.92	29.833				
11 Aug 5:07 pm CDT	93	66	41	NNE	16	10.00		FEW040 SCT075 BKN130 BKN250		29.92	29.833				
11 Aug 4:53 pm CDT	95	68	42	NNE	18	10.00	TS	SCT040CB SCT080 BKN170 BKN250	1012.6	29.91	29.823				
11 Aug 4:43 pm CDT	97	68	39	NNE	22	10.00	TS	SCT040CB SCT075 BKN100 BKN250		29.91	29.823				
11 Aug 4:31 pm CDT	100	70	38	NNE	20	10.00	TS	FEW040CB SCT075 BKN250		29.91	29.823				
11 Aug 3:53 pm CDT	104	68	32	VRBL	3	10.00		FEW065 BKN250	1012.8	29.91	29.823				
11 Aug 3:29 pm CDT	104	68	32	NE	8	10.00		FEW065 BKN250		29.92	29.833				
11 Aug 2:53 pm CDT	104	70	34	VRBL	3	10.00		FEW065 SCT250	1014.0	29.95	29.863				
11 Aug 1:53 pm CDT	101	70	37	CALM		10.00		FEW060 SCT250	1014.7	29.97	29.883				
11 Aug 12:53 pm CDT	101	70	37	VRBL	3	10.00		FEW250	1015.4	29.99	29.903				101
11 Aug 11:53 am CDT	98	67	37	VRBL	5	10.00		FEW250	1015.9	30.01	29.922				
11 Aug 10:53 am CDT	93	67	43	NW	8	10.00		FEW250	1016.2	30.01	29.922				

KHOU Observations on Aug 11

Weather Conditions for:

Houston, Houston Hobby Airport, TX. [KHOU](#) (NWS/FAA)

Elev: 46 ft; Latitude: 29.63750; Longitude: -95.28250

NOTE: Lost Power after 5:53 PM CDT

Time	Temp.	Dew	Relative	Wind	Wind	Visibility	WX	Clouds	Sea Level	Altimeter	Station	Precip	Precip	Precip	Precip	6 Hr	6 Hr	24 Hr	24 Hr	Quality
(CDT)	(f)	(f)	(%)	Direction	Speed	(miles)			Pressure	Setting	Pressure	1 hour	3 hour	6 hour	24 hour	Max	Min	Max	Min	Control
					(mph)				(mb)	(inches)	(inches)	(inches)	(inches)	(inches)	(inches)	Temp	Temp	Temp	Temp	
11 Aug 5:53 pm CDT	97	74	48	S	12	10.00	TS	FEW040CB SCT070 BKN250	1013.0	29.90	29.859									OK
11 Aug 5:36 pm CDT	99	73	45	S	14	10.00	TS	FEW040CB SCT070 BKN250		29.91	29.869									OK
11 Aug 4:53 pm CDT	103	67	32	VRBL	5	10.00		FEW040 SCT070 BKN250	1012.3	29.89	29.849									OK
11 Aug 3:53 pm CDT	102	67	32	ESE	12	10.00		FEW075 BKN250	1013.2	29.91	29.869									OK
11 Aug 2:53 pm CDT	103	68	33	ENE	7	10.00		FEW080 BKN250	1014.2	29.94	29.899									OK
11 Aug 1:53 pm CDT	100	67	34	CALM		10.00		SCT250	1015.0	29.96	29.919									OK
11 Aug 12:53 pm CDT	99	68	37	VRBL	3	10.00		SCT250	1015.7	29.98	29.939					99	80			OK
11 Aug 11:53 am CDT	95	66	39	NW	6	10.00		SCT250	1016.2	30.00	29.959									OK
11 Aug 10:53 am CDT	92	70	49	NNW	8	10.00		SCT250	1016.3	30.00	29.959									OK

KSGR Observations on Aug 11

Weather Conditions for:

Houston, Sugar Land Municipal / Hull Field Airport, TX. **KSGR** (NWS/FAA)

Elev: 82 ft; Latitude: 29.62222; Longitude: -95.65667

Time	Temp.	Dew	Relative	Wind	Wind	Visibility	WX	Clouds	Sea Level	Altimeter	Station	Precip	Precip	Precip	6 Hr
(CDT)	(f)	(f)	(%)	Point	Direction	Speed			Pressure	Setting	Pressure	1 hour	3 hour	6 hour	Max
						(mph)	(miles)		(mb)	(inches)	(inches)	(inches)	(inches)	(inches)	Temp
11 Aug 8:45 pm CDT	76	73	91	S	7	10.00	-RA	CLR		30.00	29.920	T			
11 Aug 8:30 pm CDT	76	72	88	SSE	12	10.00	VCTS -RA	CLR		29.99	29.910	T			
11 Aug 8:18 pm CDT	76	72	88	SSE	14	10.00	-RA	CLR		29.99	29.910	T			
11 Aug 8:03 pm CDT	76	71	85	SE	8	10.00	VCTS -RA	CLR		29.99	29.910	T			
11 Aug 7:53 pm CDT	75	71	88	SSW	14	10.00	-TSRA	CLR	1015.1	29.97	29.890	0.30			
11 Aug 7:23 pm CDT	75	71	88	WSW	18G31	3.00	+TSRA BR	FEW050 BKN090 BKN110		30.00	29.920	0.27			
11 Aug 7:05 pm CDT	77	72	85	SSE	18G25	1.50	+TSRA	SCT034 BKN080 OVC090		30.01	29.930	0.11			
11 Aug 6:53 pm CDT	78	71	79	SSE	21G28	1.75	+TSRA	BKN026 BKN050 OVC085	1016.7	30.02	29.940	0.59		0.59	104
11 Aug 6:45 pm CDT	80	68	67	ENE	8G51	1.00	+TSRA	BKN026 BKN050 OVC080		30.01	29.930	0.39			
11 Aug 6:39 pm CDT	78	68	71	WNW	36G51	0.50	+TSRA	BKN022 BKN050 OVC075		29.97	29.890	0.36			
11 Aug 6:37 pm CDT	78	69	74	WNW	26G61	0.75	+TSRA	BKN020 BKN050 OVC090		29.97	29.890	0.33			
11 Aug 6:31 pm CDT	77	70	79	N	26G61	1.00	+TSRA SQ	SCT020 BKN050 BKN075		29.98	29.900	0.14			
11 Aug 6:12 pm CDT	97	75	50	SSE	16	10.00	TS	SCT090		29.92	29.840				
11 Aug 6:01 pm CDT	98	74	46	S	20G25	10.00	VCTS	FEW003 FEW090		29.89	29.810				
11 Aug 5:53 pm CDT	99	73	44	SSE	20	10.00		SCT090	1012.5	29.90	29.820				
11 Aug 4:53 pm CDT	103	66	31	CALM		10.00		SCT085	1012.1	29.89	29.810				
11 Aug 3:53 pm CDT	103	67	32	E	6	10.00		CLR	1013.1	29.92	29.840				
11 Aug 2:53 pm CDT	103	68	33	VRBL	5	10.00		CLR	1014.2	29.95	29.870				
11 Aug 1:53 pm CDT	100	68	36	VRBL	3	10.00		CLR	1014.9	29.97	29.890				
11 Aug 12:53 pm CDT	99	67	36	NNW	7	10.00		CLR	1015.6	29.99	29.910				99
11 Aug 11:53 am CDT	95	68	42	NW	6	10.00		CLR	1016.1	30.01	29.930				
11 Aug 10:53 am CDT	92	71	51	NNW	8	10.00		CLR	1016.1	30.01	29.930				
11 Aug 9:53 am CDT	87	75	68	NW	7	10.00		CLR	1016.5	30.02	29.940				